

connect_pg_net

NAME

`connect_pg_net`
Connects power, ground, and tie-off pins to the power and ground nets. The command supports both an automatic mode and a manual mode. The default is automatic mode. In automatic mode, the tool derives the power and ground nets from the power intent of the design. Automatic mode requires committed UPF descriptions and logic libraries with power and ground pins. In manual mode, you specify the power and ground nets.

SYNTAX

```
status connect_pg_net
  [-automatic]
  [-pg]
  [-tie]
  [-all_blocks]
  [-block block_list]
  [-design design]
  [-net net]
  [port_pin_list]
  [-create_nets_only]
  [-reconnect true|false]
  [-report_connection_summary true|false]
  [-verbose]
```

Data Types

block_list	list
design	collection
net	list
port_pin_list	list

ARGUMENTS

- `-automatic`
Specifies automatic mode. This option cannot be used with `-net` or `port_pin_list` options.
- `-pg`
Specifies that only leaf power and ground pins are connected. If `-pg` and `-tie` options are not specified, both PG and tie-off pins are connected in automatic mode. This option cannot be used with `-create_nets_only`, `-net` or `port_pin_list` options.
- `-tie`
Specifies that only tie-off pins are connected. If `-pg` and `-tie` are not specified, both PG and tie-off pins are connected in automatic mode. This option cannot be used with `-create_nets_only`, `-net` or `port_pin_list` options.
- `-all_blocks`
Traverses all physical blocks during auto PG connection. This option cannot be used with `-block`, `-net` or `port_pin_list` options.
- `-block block_list`
Traverses the current design, as well as the physical blocks specified in `block_list` during auto PG connection. Blocks in `block_list` that are not in current design are ignored. This option cannot be used with `-all_blocks`, `-net` or `port_pin_list` options.
- `-design design`
Specifies the design in which to connect power and ground nets. If no design is specified, the power and ground nets are connected in the current design.
- `-net net`
Specifies the PG net to connect. The net argument can be a name or collection. The net must be a scalar (single-bit) PG net and must exist in the current design.
- `port_pin_list`
Specifies a list of ports and pins to which the net is to be connected. The ports and pins must exist in the current design.

By default, if a specified port or pin is already connected, the command disconnects the existing connection and makes a new connection. The behavior of reconnection can be specified by

-reconnect command option.

- create_nets_only
Creates topmost PG nets automatically based on power constraints, even when the design contains unmapped cells.
- reconnect true|false
Specifies whether to allow reconnection in manual mode when a specified port/pin already has an existing net connection. By default, reconnection is allowed.
- report_connection_summary true|false
Specifies whether to report summary of PG connection changes. By default, pg connection summary is reported.
- verbose
Prints the details of the changes to the power, ground, and tie-off connections.

DESCRIPTION

The `connect_pg_net` command creates the power and ground network for your design by connecting the power, ground, and tie-off pins to the power and ground nets in your design.

The command supports two modes: automatic mode and manual mode.

In automatic mode, the tool derives the power and ground nets from the power intent of the design. UPF must be committed before running `connect_pg_net` in automatic mode. In addition, automatic mode requires logic libraries with power and ground pins. In automatic mode, the tool creates the power and ground nets if they do not already exist. The tool reports PG net and PG pin connections status at the end.

For non UPF designs, automatic mode derives power and ground nets based on the primary supply nets of default power domain. The system default values of the application options (`mv.pg.default_power_supply_net_name`, `mv.pg.default_power_supply_port_name`, `mv.pg.default_ground_supply_net_name` and `mv.pg.default_ground_supply_port_name`) are used to specify the primary supply names of the default power domain if no PG nets from Verilog or DEF exist in the non UPF design. If the application options are not explicitly set and one power net and one ground net is specified as PG nets in Verilog or DEF, automatic mode automatically chooses them as primary supply names of the design. If there is more than one power or ground nets, the application option has to be explicitly set to specify the primary supply names, even if it is the same as the application option's system default.

In manual mode, use `-net` option to specify the power and ground nets. Using `-net` option invokes the tool in manual mode. You can use manual mode for single-voltage or multivoltage designs.

EXAMPLES

The following example uses the `connect_pg_net` command in automatic mode to connect all unconnected power and ground pins based on the power domain connections.

```
prompt> connect_pg_net -automatic
```

```
*****
Report : Power/Ground Connection Summary
Design : TOP
Version: N-2017.09-SP6-VAL
Date   : Tue Jun 19 02:45:44 2018
*****
```

PG net name	PG pin count(previous/current)
Power net VDD1SD	0/8
Power net VDD3	0/6
Power net VDD2	0/6
Power net VDD1	0/0
Unconnected nwell pins	20
Ground net VSS	0/20
Unconnected pwell pins	20

Information: connections of 40 power or ground pins are created or changed.

The following example uses the `connect_pg_net` command in manual mode to connect power pin `vdd_io` on cell `I_PAD1` to power net `VDD_IO`. The command also connects power pin `vdd_core` on the same cell to power net `VDD_CORE`.

```
prompt> connect_pg_net -net VDD_IO [get_pins I_PAD1/vdd_io]
```

```
prompt> connect_pg_net -net VDD_CORE [get_pins I_PAD1/vdd_core]
```

SEE ALSO

[resolve_pg_nets\(2\)](#)

[commit_upf\(2\)](#)

[mv.pg.default_power_supply_net_name\(3\)](#)

[mv.pg.default_power_supply_port_name\(3\)](#)

[mv.pg.default_ground_supply_net_name\(3\)](#)

[mv.pg.default_ground_supply_port_name\(3\)](#)

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